Adam air flight 574

Flight model: Boeing 737
Date: January 01, 2007

From: Surabaya, Indonesia
To: Manado, Indonesia

Adam air is a low cost carrier.

Problem

After takeoff, the plane experiences some turbulence while the autopilot is turned on. After sometime, the ATC notices the flight has flew off course and is heading into violent storm. In the plane, the pilots try to find how far they have drifted away from the course. The plane has to passes through the storm to correct the off course. When the navigation instrument does not work properly in middle of the sea it becomes a horrible nightmare for the pilots. The pilots ask their location from the ATC. Then, there is a warning of auto pilots disengage followed by many instruments going blank. The plane starts to bank at a dangerous angle. The pilots try to re-engage the navigation system. Now, the nose started to pitch downwards. The the **sink rate** alarm pops up which indicate they are entering into a steep decent. finally, crashes.

Investigation

The wreckage found in the deep sea tells how the plane was ripped on the violent crash. After a crash, the DVR and FDR are stored in the fresh water till it is taken to the lab so that the electronics do not rust. Keeping them submerged delays the rusting process. To get the location details of a plane, they use **IRS (Inertial referral system)**, it tracks every adjustments made on the planes course. It then feed on to the autopilot system. The IRS provides the data on heading, altitude, speed etc. The IRS has to be calibrated before take off which includes giving the source of the journey. Usually the gate number is given as the starting point to track. And, the data from the FDR confirms the IRS was calibrated before take off. The maintenance record show it had numerous record on the problem in the IRS system. The DVR, show IRS is not sole reason for the plane to crash. The FDR and DVR, show the pilot switch the manual navigation when the initial sign of trouble was recognized in the IRS. When switching from one mode to another usually the instruments usually go blank for sometime for around 30 seconds. One of the system is **ADI (Artificial horizon)**, which tell if the plane is flying in a straight level. This is when the plane starts to bank. The pilot should have flown in a straight level when the mode were switched which the pilots failed to do. The plane had a natural tendency to roll to the right and not always go straight. When the auto pilot was turned on the aileron used to compensate the turn but, when the autopilot disengaged the ailerons went into neutral position. Which made it to bank right. The pilots were spatially disoriented by the thick clouds. The bank angle warning pops up when the plane rolls over 35 degrees in the 737. The pilot tries to make some correction but again gets distracted to the IRS. Usually when one pilot deals with the issue the other pilot must fly the plane, but in this case both the pilots are focused on solving the IRS problem and forgets the drive the plane. When the plane banks and nose pitches down, the pilots should have first leveled his plane and then should have pulled the nose up but, the pilot does the opposite. This makes the situation much worse. When the dive increases the plane speed also increases which is beyond the threshold the plane is designed to withstand. Which made the plane rip apart in mid air. The airline seems to not have given adequate training to its pilots on many failure situation.

Changes that were brought after the incident

Adam air were found to lack proper training. Which made the Indonesian government to revoke the Adam air license which was made after another accident occurred in its fleet. It was found that many crashes that occurred in Indonesia involved low cost carrier. This forces the government to take action on airline safety and had to bring reforms.

Reference

1. What Happened to Adam Air 574? - Mayday: Air Disaster
   `https://www.youtube.com/watch?v=RdpjsOQ8BLU`